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**HK IMO Training 2018-2019**  
**Problem Set 5**  
**Date due: 27 October 2018**

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**Problem 21.** There is a pile of 1000 matches. Two players each take turns and can take 1 to 5 matches. It is also allowed at most 10 times during the whole game to take 6 matches. For example, 7 exceptional moves can be done by the first player and 3 moves by the second and then no more exceptional moves are allowed. Whoever takes the last match wins. Determine which player has a winning strategy.

**Problem 22.** To every vertex of a regular pentagon a real number is assigned. We may perform the following operation repeatedly: we choose two adjacent vertices of the pentagon and replace each of the two numbers assigned to these vertices by their arithmetic mean. Is it always possible to obtain the position in which all five numbers are zeroes, given that in the initial position the sum of all five numbers is equal to zero?

**Problem 23.** Triangle  $ABC$  is given. Let  $M$  be the midpoint of the segment  $AB$  and  $T$  be the midpoint of arc  $BC$  not containing  $A$  of the circumcircle of  $\triangle ABC$ . The point  $K$  inside  $\triangle ABC$  is such that  $MATK$  is an isosceles trapezoid with  $AT \parallel MK$ . Show that  $AK = KC$ .

**Problem 24.** For a positive integer  $k$ , let  $d(k)$  denote the number of divisors of  $k$  (e.g.  $d(12) = 6$ ) and let  $s(k)$  denote the digit sum of  $k$  (e.g.  $s(12) = 3$ ). A positive integer  $n$  is said to be *amusing* if there exists a positive integer  $k$  such that  $d(k) = s(k) = n$ . What is the smallest amusing odd integer greater than 1?